



MISSOURI

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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 4.4

Coordinate Systems

Coordinates

Suppose $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for a vector space V and $\mathbf{x}$ is a vector in V . The coordinates of $\mathbf{x}$ relative to the basis B (or the B -coordinates of $\mathbf{x}$) are the weights $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$

The B -coordinate vector of $\mathbf{x}$ is

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

Example 1

The sets

$$B_1 = \{1, t, t^2\}$$

and

$$B_2 = \{1, 1 + t, 1 + t + t^2\}$$

are both bases for $\mathbb{P}_2$. Find the coordinate vectors for

$$\mathbf{p}(t) = 4 - 6t + 3t^2$$

under the basis B_1 and under the basis B_2 .

Example 2

The set

$$B = \left\{ \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^2$. Find the vector $\mathbf{x}$ determined by the B -coordinate vector

$$[\mathbf{x}]_B = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$$

Note that, unless otherwise specified, $\mathbf{x}$ is a vector relative to the standard basis.

The Unique Representation Theorem

Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . Then, for each $\mathbf{x}$ in V , there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$

Example 3

Prove the Unique Representation Theorem

Example 4

The set

$$B = \left\{ \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^2$. Find the coordinate vector

$$[\mathbf{x}]_B \text{ of } \mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Change of Coordinates Matrices

To change coordinates from a basis B to the standard basis in $\mathbb{R}^n$, we can compute

$$\mathbf{x} = P_B [\mathbf{x}]_B$$

for some matrix P_B . In general, we call P_B the change of coordinates matrix from B to the standard basis in $\mathbb{R}^n$.

Similarly, we can go the other direction using the inverse of P_B .

$$P_B^{-1} \mathbf{x} = [\mathbf{x}]_B$$

Coordinate Mappings

Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . This vector space V might not be $\mathbb{R}^n$.

Then, the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_B$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

Isomorphisms

A one-to-one linear transformation from a vector space V onto a vector space W is called an isomorphism from V onto W .

If there exists an isomorphism from V onto W , then V and W are said to be isomorphic.

Two isomorphic vector spaces have the same general structure, and every calculation in one space is accurately reproduced in the other.

The coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_B$ is an isomorphism.

Example 5

Recall that $B = \{1, t, t^2, t^3\}$ is the standard basis for the vector space $\mathbb{P}_3$.

Elements in $\mathbb{P}_3$ are of the form

$$\mathbf{p}(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

Thus, the coordinate vector of $\mathbf{p}$ relative to B is

$$[\mathbf{p}]_B = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

This is an isomorphism from $\mathbb{P}_3$ onto $\mathbb{R}^4$.

Example 6

Use coordinate vectors to test the linear independence of the set of polynomials.

$$\{1 + 2t^3, 2 + t - 3t^2, -t + 2t^2 - t^3\}$$

Do these polynomials form a basis for $\mathbb{P}_3$?
